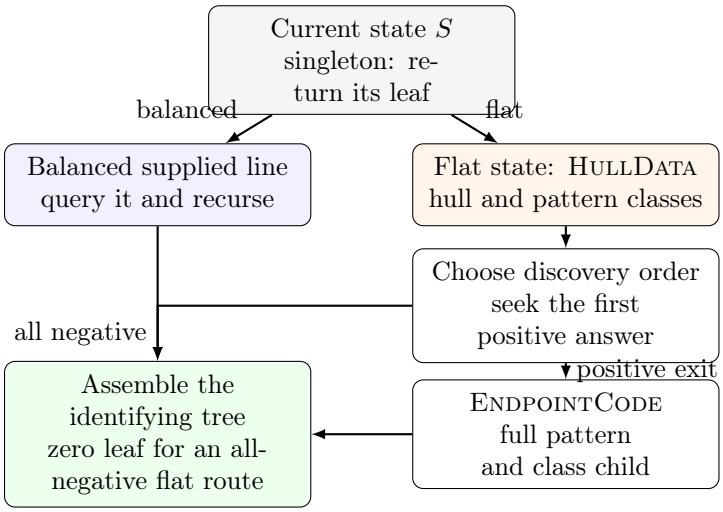


Figure 1: The common recursive structure. Arrows describe the output of one stage passed to the next. The maximum-depth routines construct class children before choosing the schedule, so their stored heights can set deadlines; expected-depth routines use original prior masses. The flat stage has an all-negative leaf only when the zero pattern is present. Scheduling and decoding use supplied hull lines.

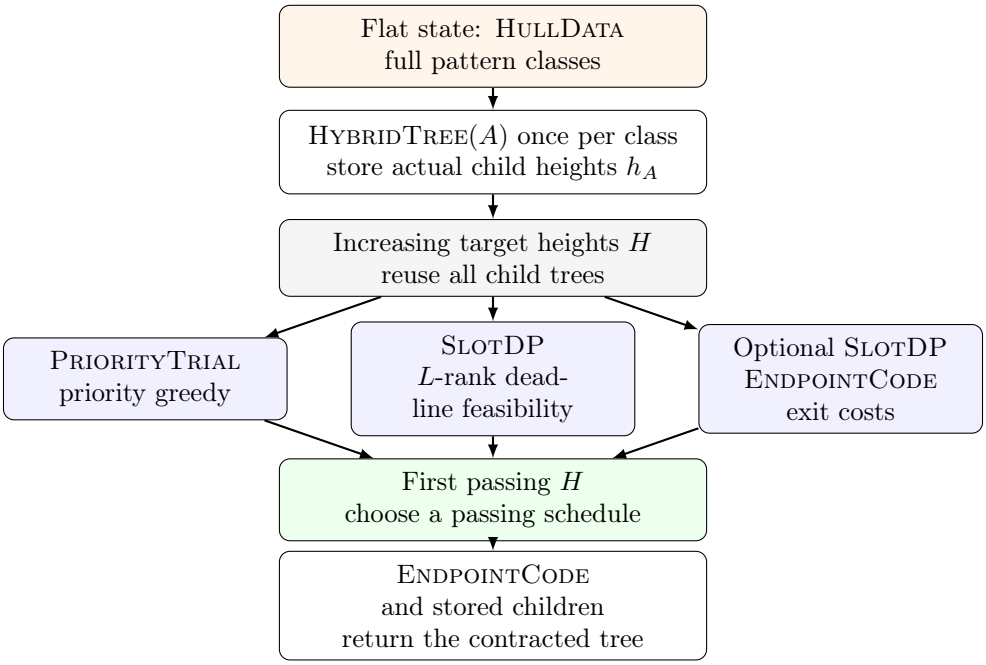


Figure 2: Construction of the polynomial maximum-depth tree at a flat state. Arrows pass explicitly computed geometry, heights, or schedules. The logarithmic cap L is fixed at the original root. The priority alternative guarantees that some height trial passes; the exact and whole-exit alternatives can accept an earlier trial. The unknown optimum is used only in the proof, never in this procedure.

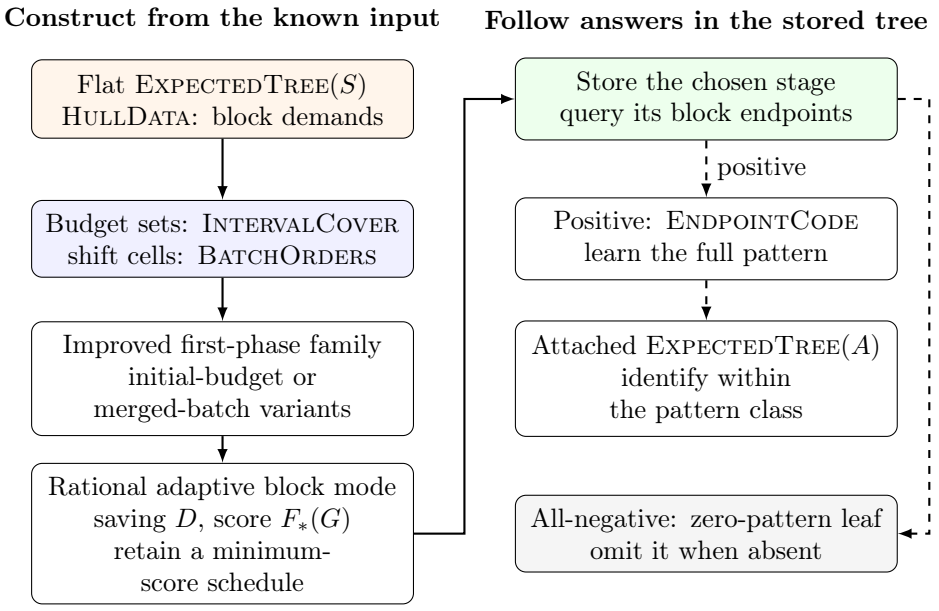


Figure 3: The improved polynomial expected-depth procedure. Solid arrows are construction-time data flow, computed from the supplied incidence table and rational priors. Dashed arrows show how the unknown hypothesis traverses the resulting tree; endpoint decoders and recursive children are already constructed. Adaptive modes choose whether the first positive block stops early or queries both endpoints. The certificate's mixture is used in the proof of the bound, not to randomize execution.

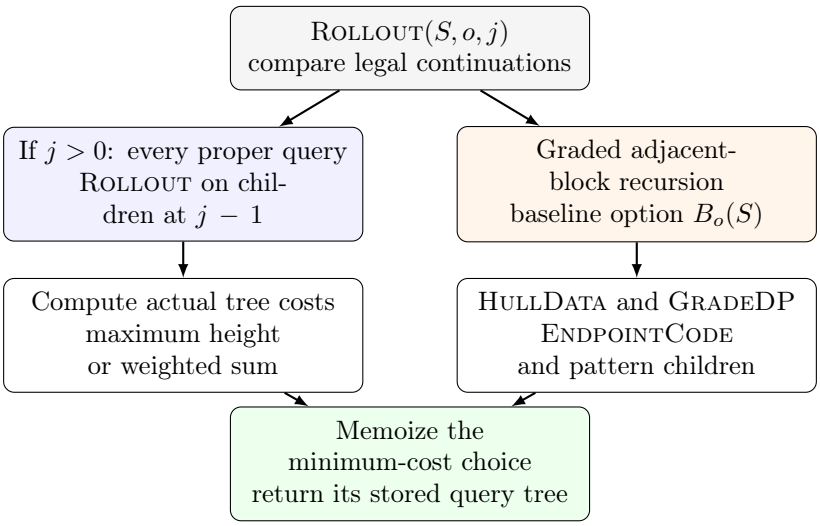


Figure 4: Exact lookahead wrapped around the quasipolynomial baseline. The arrows denote calls and the returned trees or costs. The baseline may end lookahead early at any reached state. The logarithmic cap on j bounds the number of reachable transcripts; it does not truncate the completed identification tree. The proof compares this finite search with an optimal tree, but the search evaluates only constructed trees.